

Interior Designer App - Technical Report

Project: My Interior Designer App

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Executive Summary

This is an AI-powered interior design application that enables users to upload photos of empty rooms and generate redesigned interior images using artificial intelligence. The application features 3D visualization capabilities, furniture measurements, and support for 10+ room types with phone number authentication.

Technology Stack

Frontend Technologies

Core Framework & Build Tools

- **React** (v18.3.1) - Primary UI framework
- **React DOM** (v18.3.1) - React rendering for web
- **Vite** (v5.4.19) - Fast build tool and development server
- **@vitejs/plugin-react** (v5.0.2) - Official Vite plugin for React

Routing

- **React Router DOM** (v7.8.2) - Client-side routing and navigation

3D Visualization

- **Three.js** (v0.159.0) - 3D graphics library
- **@react-three/fiber** (v8.18.0) - React renderer for Three.js
- **@react-three/drei** (v9.122.0) - Helper components for react-three-fiber

Styling

- **Tailwind CSS** (v4.1.12) - Utility-first CSS framework
- **PostCSS** (v8.5.6) - CSS transformation tool
- **Autoprefixer** (v10.4.21) - PostCSS plugin for vendor prefixes
- **@tailwindcss/postcss** (v4.1.13) - Tailwind PostCSS integration

Custom Design System:

- Custom color palettes (brand, gold)
- Custom animations (float, fadeup)
- Soft shadows and card effects
- Inter font family integration
- Dark mode support via class strategy

Backend Technologies

Server Framework

- **Express.js** (v4.19.2) - Web application framework
- **Node.js** - JavaScript runtime environment
- **CORS** (v2.8.5) - Cross-origin resource sharing middleware

File Handling

- **Multer** (v1.4.5-lts.1) - Multipart/form-data handling for file uploads
- **Formidable** (v3.5.4) - Form and file upload parsing
- **Canvas** (v3.2.0) - Image processing capabilities

Environment & Configuration

- **dotenv** (v16.4.5) - Environment variable management
- **Concurrently** (v8.2.2) - Run multiple commands simultaneously

AI & Machine Learning Services

Frontend AI Integration

- **@google/generative-ai** (v0.24.1) - Google Gemini API client
- **@huggingface/inference** (v4.7.1) - Hugging Face API integration
- **OpenAI** (v5.19.1) - OpenAI API client

Backend AI Services

- **Replicate** (v0.31.0) - AI model deployment platform
- **OpenRouter API** - AI model routing service
- **Node-fetch** (v3.3.2) - HTTP client for API requests

Python ML Stack (ml_models/)

- **TensorFlow** (v2.15.0) - Deep learning framework
- **PyTorch** (v2.1.0) - Machine learning framework
- **Torchvision** (v0.16.0) - Computer vision models
- **NumPy** (v1.24.3) - Numerical computing
- **Pillow** (v10.0.1) - Image processing
- **OpenCV** (v4.8.1.78) - Computer vision library
- **Scikit-learn** (v1.3.0) - Machine learning utilities
- **Matplotlib** (v3.7.2) - Data visualization
- **Trimesh** (v4.0.5) - 3D mesh processing
- **Open3D** (v0.18.0) - 3D data processing
- **Flask** (v2.3.3) - Python web framework
- **Flask-CORS** (v4.0.0) - CORS support for Flask
- **Requests** (v2.31.0) - HTTP library

Authentication & Backend Services

Firebase Integration

- **Firebase** (v12.2.1) - Complete backend solution
 - Firebase Authentication (Phone number sign-in enabled)
 - Firebase Firestore (NoSQL database)
 - Firebase Storage (File storage)

Deployment & Hosting

- **Vercel** - Primary deployment platform
 - Serverless functions
 - Static site hosting
 - Automatic deployments
 - Environment variable management
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Project Architecture

Frontend Structure

React Components (src/)

- **App.jsx** - Main application component
- **main.jsx** - Application entry point
- **ErrorBoundary.jsx** - Error handling wrapper
- **FirebaseDebug.jsx** - Firebase debugging utilities
- **Footer.jsx** - Application footer
- **Header.jsx** - Application header
- **PanoramaViewer.jsx** - 360° room view component
- **ProtectedRoute.jsx** - Authentication guard
- **RoomCard.jsx** - Room display card
- **sidebar.jsx** - Navigation sidebar
- **ThreeDViewer.jsx** - 3D room visualization

Pages (src/pages/)

- **About.jsx** - About page
- **DemoLogin.jsx** - Demo authentication
- **Design.jsx** - Design generation interface
- **Home.jsx** - Landing page
- **Login.jsx** - Authentication page
- **RoomDetails.jsx** - Room detail view

Context Providers

- **AuthContext.jsx** - Authentication state management

Backend Structure

API Server (server.js)

- Express server on port 8787
- File upload handling
- AI image generation orchestration
- Multiple AI provider integration (Replicate, OpenRouter)
- Furniture layout generation
- Room design description generation

API Routes

- `/api/health` - Health check endpoint
- `/api/design.js` - Design generation endpoint
- Proxy configuration for frontend-backend communication

Machine Learning Module (ml_models/)

- **ml_api.py** - ML model API server
- **3d_model_generator.py** - 3D model generation
- **model_architecture.py** - Neural network architectures
- **train_model.py** - Model training pipeline
- **quick_start.py** - Quick start script
- **setup_ml.py** - ML environment setup
- **test_3d_generation.py** - 3D generation testing

- **training_data/** - Training data directory

Configuration Files

- **vite.config.mjs** - Vite configuration with API proxy
 - **tailwind.config.js** - Tailwind CSS customization
 - **postcss.config.js** - PostCSS configuration
 - **vercel.json** - Vercel deployment configuration
 - **package.json** - Node.js dependencies and scripts
 - **.env** - Environment variables (API keys)
 - **replicate.env** - Replicate API token
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Key Features

Room Design Generation

- Upload empty room photos
- Support for 10+ room types (living room, bedroom, kitchen, office, etc.)
- Custom element addition (TV, bookshelf, piano, etc.)
- Multiple design styles (modern, traditional, minimalist, industrial)
- AI-powered furniture placement with measurements

3D Visualization

- Interactive 3D room preview
- Three.js-based rendering
- React Three Fiber integration
- Panoramic 360° room views
- Real-time 3D model generation

Authentication

- Firebase Authentication
- **Phone number sign-in** (primary method)
- Protected routes
- Demo login capability
- Session management

AI Integration

- Multiple AI provider support
 - Google Gemini API for text generation
 - Replicate for image generation
 - OpenRouter for model routing
 - Hugging Face model integration
 - Custom ML models with TensorFlow/PyTorch
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Development Scripts

```
"dev": "vite --host --port 5175 --open"
"api": "node server.js"
"dev:full": "concurrently \"npm run api\" \"npm run dev\""
"dev:ml": "python start_with_ml.py"
```

```
"ml:setup": "cd ml_models && python setup_ml.py"
"ml:train": "cd ml_models && python train_model.py"
"ml:api": "cd ml_models && python ml_api.py"
"ml:test": "cd ml_models && python test_3d_generation.py"
"ml:start": "cd ml_models && python quick_start.py"
"test:api": "node test_ml_api.js"
"build": "vite build"
"preview": "vite preview"
"vercel-build": "vite build"
```

API Integrations

External APIs

1. **Google AI Studio** - Gemini API for AI-powered design suggestions
2. **Replicate** - AI model deployment and image generation
3. **OpenRouter** - AI model routing (Flux Pro 1.1)
4. **Hugging Face** - Pre-trained model inference
5. **Firebase** - Authentication and database services
6. **Unsplash** - Placeholder design images

Development Environment

Prerequisites

- Node.js (with npm)
- Python 3.x (for ML models)
- Google AI Studio account
- Replicate API account
- Firebase project

Port Configuration

- Frontend: localhost:5175
- Backend API: localhost:8787
- ML API: Configurable in ml_models

Data Flow

1. **User uploads room photo** → Frontend (React)
2. **Image sent to backend** → Express server (Multer)
3. **AI processing** → Multiple AI providers (Replicate/OpenRouter/Gemini)
4. **3D generation** → Python ML models (TensorFlow/PyTorch)
5. **Visualization** → Three.js rendering
6. **Results stored** → Firebase Firestore
7. **User authentication** → Firebase Auth (Phone number)

Documentation Files

- **README.md** - Main project documentation
- **ARCHITECTURE.md** - System architecture details

- **3D_FEATURE_GUIDE.md** - 3D visualization guide
 - **3D_SYSTEM_README.md** - 3D system documentation
 - **3D_VIEW_FEATURE.md** - 3D view feature specifications
 - **ml_models/README.md** - ML module documentation
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Build & Deployment

Production Build

- Vite builds optimized production bundle
- Assets compiled to `dist/` directory
- Code splitting and tree shaking enabled
- CSS minification with Tailwind

Deployment Platform

- **Vercel** serverless deployment
 - Automatic Git deployments
 - Environment variables configured
 - Static file serving from CDN
 - Serverless API functions
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Security Considerations

- API keys stored in `.env` files (not committed to Git)
 - CORS configured for specific origins
 - Protected routes with authentication
 - Firebase security rules
 - Environment variable validation
 - Secure file upload handling
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Image Sources

Design Generation

- Replicate AI models for custom generation
 - Unsplash API for fallback images
 - Custom ML-generated 3D models
 - Cached design templates
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Future Enhancements

Based on the current architecture, potential areas for expansion:

- Additional AI model providers
 - Advanced 3D rendering capabilities
 - Real-time collaboration features
 - Extended room type support
 - Enhanced furniture catalog
 - AR visualization integration
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Conclusion

This Interior Designer App represents a comprehensive full-stack application combining modern web technologies (React, Vite, Tailwind) with advanced AI/ML capabilities (TensorFlow, PyTorch, multiple AI APIs) and 3D visualization (Three.js). The application demonstrates expertise in:

- Modern frontend development
- Backend API design
- Machine learning integration
- 3D graphics programming
- Cloud deployment
- Authentication systems
- Real-time data processing

Total Dependencies: 30+ npm packages, 14 Python packages

Total Components: 18 React components

API Integrations: 5+ external services

Documentation Pages: 5+ markdown files

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